Clinical Study Report (CSR) — SAMPLE_STUDY

Title: Sample Validation Study for Condition X Study ID: SAMPLE-001 Version: 0.1-draft

Introduction This CSR documents a minimal reproducible validation exercise using a de■identified synthetic dataset included in the repository. This draft uses the available repository evidence (see Traceability) and is intended to demonstrate the structure and content for a real CSR.

Objectives Primary objective: To evaluate the diagnostic performance of the algorithm for detecting Condition X as measured by sensitivity and specificity.

Methods 3.1 Study design Retrospective, single■arm, dataset-based validation using an internally available de■identified dataset located at data/sample_study/sample.csv. Dataset contains N=425 records (125 positives, 300 negatives) as prepared for reproducible validation testing.

3.2 Population - Inclusion: records labeled as positive or negative for Condition X in data/sample_study/sample.csv. - Exclusion: none applied for this synthetic dataset.

3.3 Statistical Analysis Plan (SAP) - No separate SAP file was provided in the repository. For this draft, the analysis follows the simple SAP outlined below and implemented in analysis/analysis.py: - Primary metrics: sensitivity and specificity with 95% Wilson confidence intervals. - Confusion matrix (TP, FP, TN, FN) will be reported. - ROC and AUC will be plotted and included in the evidence package. - NOTE: Replace this section with the official SAP and version-controlled file when available; annotate SAP version and commit SHA in the final CSR.

3.4 Analysis methods (reproducible) - Analysis script used: analysis/analysis.py (computes sensitivity/specificity and Wilson CIs)
- Plotting: analysis/plot_metrics.py (produces ROC and confusion matrix in analysis/results/plot.png)
- All outputs are reproducible by running python3 analysis/analysis.py data/sample_study/sample.csv.

Results 4.1 Flow diagram and dataset Total records: 425 Positives: 125, Negatives: 300

4.2 Primary endpoint results (see analysis/results/metrics.json and plot) - True positives (TP): 124
- True negatives (TN): 299
- False positives (FP): 1
- False negatives (FN): 1

Sensitivity: 0.992 (95% Wilson CI: 0.956074 – 0.998586) Specificity: 0.996667 (95% Wilson CI: 0.981363 – 0.999411)

Figures and tables: - Confusion matrix and ROC plot: analysis/results/plot.png (included in evidence package) - Metrics JSON: analysis/results/metrics.json

4.3 Interpretation - In this synthetic dataset, the algorithm demonstrates very high sensitivity and specificity; these findings are illustrative only and must be confirmed using a properly designed clinical validation trial with the SAP, pre-specified analysis, and regulatory oversight.

Limitations The dataset used is synthetic and stored within the repository for demonstrative purposes; it does not represent an independent, prospective clinical evaluation.

No independent SAP or signed CSR currently included — replace with formal SAP and CSR signatures for submission.

Conclusion

The reproducible validation example shows algorithm performance consistent with the claims in the repository templates; however, the evidence package must be expanded with real de■identified clinical datasets and a formal SAP before submission to regulatory authorities.

Appendices Evidence package: see evidence_package.zip (generated by scripts/package_evidence.py) — includes dataset, analysis outputs, plot and manifest (SHA256 checksums). Traceability entries: regulatory/TRACEABILITY.md

Signatures: - Principal investigator: ____ **- Biostatistician:** ___ ***- Date:*** _____